```
clc;
clear;
close all;
% Input Bits
data = randi([0 1],20,1);
% Convolutional Encoder (Rate 1/2, Constraint Length = 3)
trellis = poly2trellis(3,[7 5]);
encoded = convenc(data,trellis);
% Display
disp('Input Bits:');
```

```
Input Bits:
```

```
disp(data');
```

```
     0     1     0     0     1     1     0     1     0     0     0     1     1     1     0     0     0
```

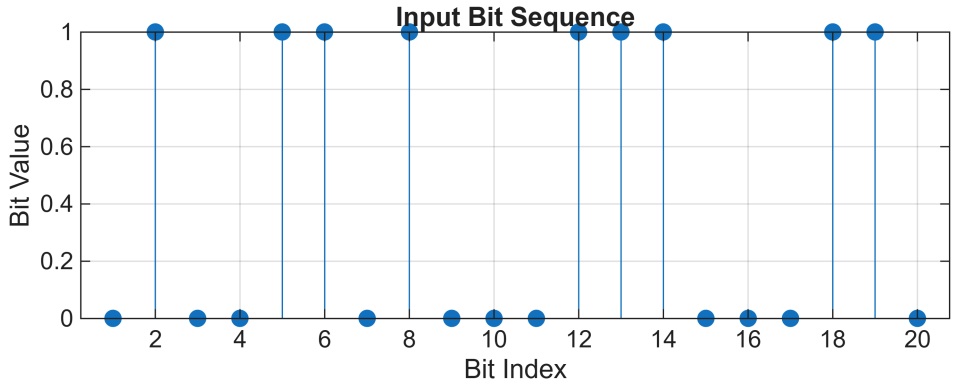
```
disp('Encoded Bits:');
```

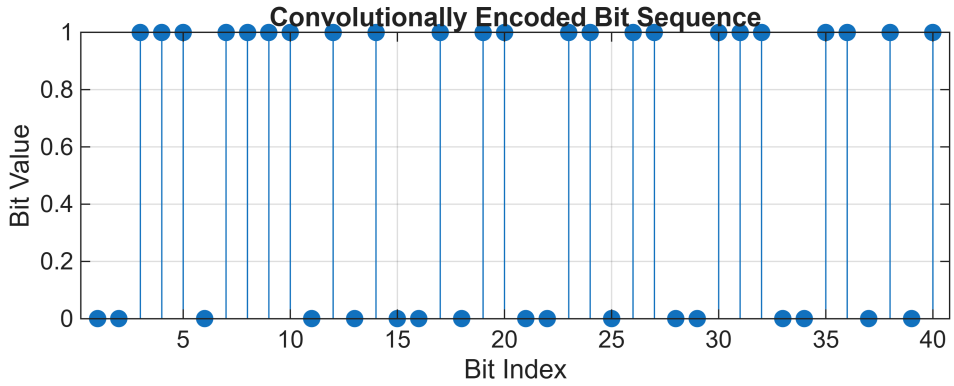
```
Encoded Bits:
```

```
disp(encoded');
```

```
     0     0     1     1     1     0     1     1     1     1     0     1     0     1     0     0     1
```

```
% Plot
figure;
subplot(2,1,1)
stem(data,'filled')
title('Input Bit Sequence')
xlabel('Bit Index')
ylabel('Bit Value')
grid on
subplot(2,1,2)
stem(encoded,'filled')
title('Convolutionally Encoded Bit Sequence')
xlabel('Bit Index')
ylabel('Bit Value')
grid on
```


Input Bit Sequence
Bit Value
Bit Index
0
0.2
0.4
0.6
0.8
1
2
4
6
8
10
12
14
16
18
20


Convolutionally Encoded Bit Sequence
Bit Value
Bit Index
0
0.2
0.4
0.6
0.8
1
5
10
15
20
25
30
35
40